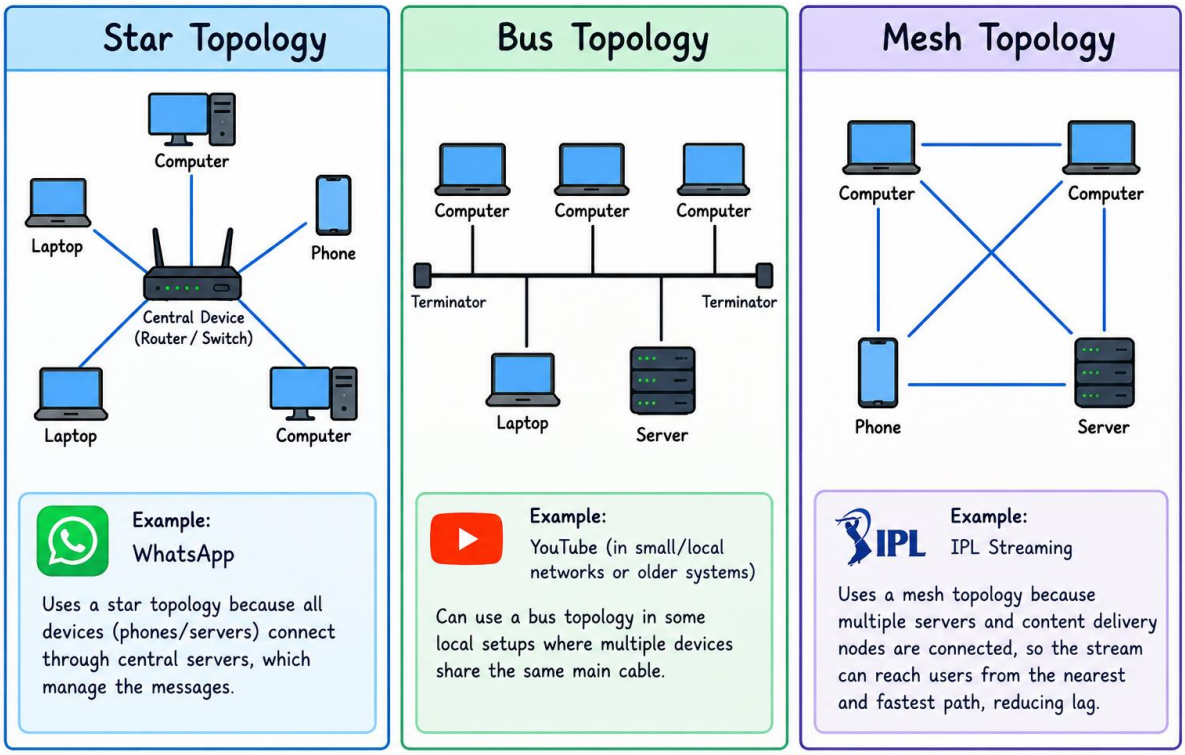
Network Topology

Task 1:- Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines



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Task 2:- Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Comparison of Network Topologies

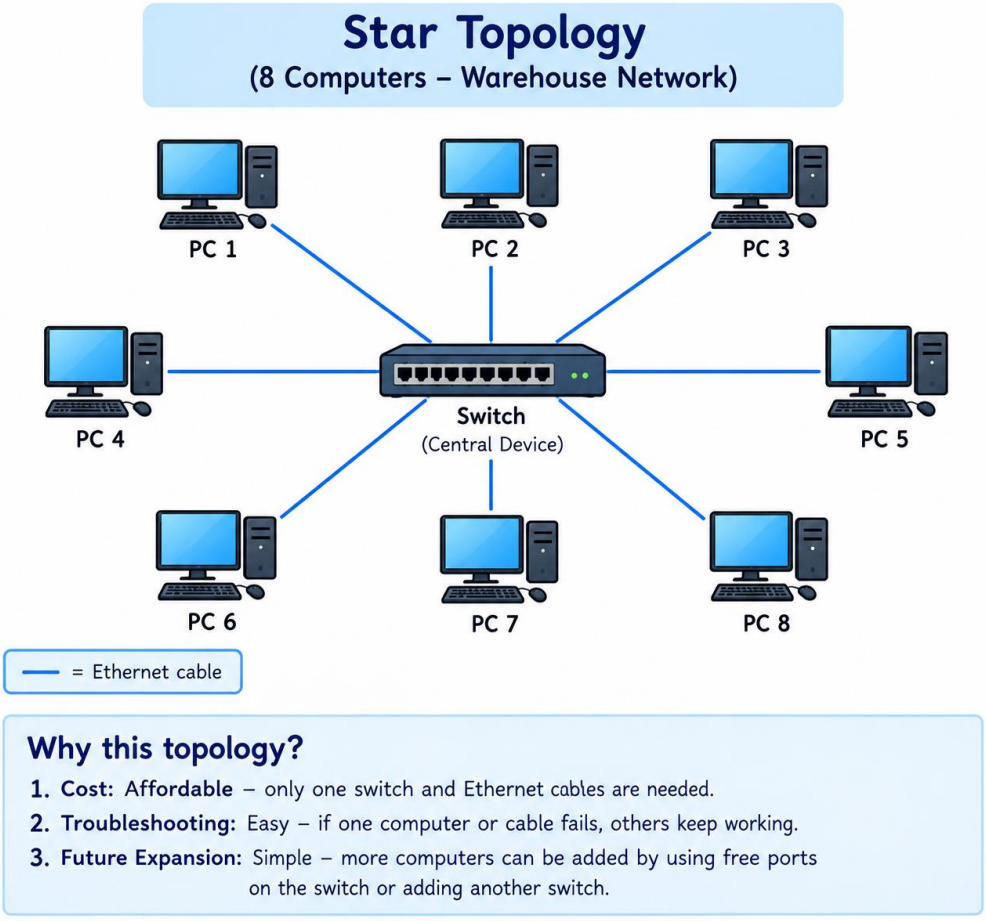
Topology	4 Pros (Advantages)	4 Cons (Disadvantages)	Best Use Case / Example
Star	1. Easy to install. 2. Easy to manage. 3. Easy to find problems. 4. One computer failure doesn't affect others.	1. Needs more cables. 2. Costs more. 3. Central switch failure stops the network. 4. Depends on the central device.	College computer lab — computers connect to one central switch.
Bus	1. Uses less cable. 2. Low cost. 3. Easy for small networks. 4. Simple design.	1. Main cable failure stops the network. 2. Hard to find problems. 3. Slow with many devices. 4. Limited cable length.	Small old computer network — devices share one main cable.
Ring	1. Data moves in order. 2. Fewer data collisions. 3. Devices get turns to send data. 4. Works well under predictable traffic.	1. One link failure may stop the network. 2. Adding devices is difficult. 3. Finding problems can be hard. 4. Data may take longer to reach its destination.	Office network — devices are connected in a circle.
Mesh	1. Very reliable. 2. Has multiple paths. 3. Can work if one link fails. 4. Data can take another route.	1. Expensive. 2. Needs more cables or links. 3. Difficult to set up. 4. Hard to maintain.	Paytm payment infrastructure — reliable networks help keep payment services available.

Task 3:- magine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

Network Topology: Mesh Topology

1. Mesh topology provides multiple paths for data, helping players connect to game servers quickly.
2. It improves reliability because data can use another path if one connection fails.
3. It can reduce lag when the game uses nearby servers and good network routes.
4. It is expensive to set up and maintain because it needs more connections and equipment.

Task 4:- Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Consider factors like cost, ease of troubleshooting, and future expansion.



Reasoning

1. Cost: One switch and Ethernet cables make it affordable for a small warehouse.
2. Troubleshooting: If one computer or cable fails, the other computers can continue working.
3. Future expansion: More computers can be added by connecting them to available switch ports or adding another switch.

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